Mini Project 12: Securities Investor Grievance Routing Service

Course: UE24AM342AA5 — ML System Design & AgentOps | Level 3 (Jackfruit) | Team: 2–3 Students

GitHub Repository: https://github.com/harishreddy2005/securities-investor-grievance-routing

1. Executive Summary & Problem Scope

The Securities and Exchange Board of India (SEBI) centralises grievances across intermediaries (brokers, mutual funds, RTAs, depositories). Traditional single-pass RAG fails when complainants express grievances using general investment vocabulary for non-securities products (such as bank fixed deposits). We built a **production-grade 2-agent LangGraph architecture with grounded RAG** that enforces data contracts and prompt injection guardrails, resolves product-regulator jurisdiction conflicts, validates statutory citations against official regulations, and outputs advisory routing determinations with timelines for human grievance officers (never disposing complaints autonomously).

2. Multi-Agent Resolution of the Section 4.1 Difficult Case

Test Scenario: *"I invested in a fixed deposit through my bank's app. The interest credited is lower than what I was promised. I want to complain to SEBI."*

- **The Retrieval Trap:** The query contains *'invested'*, *'interest'*, and *'SEBI'*, erroneously biasing naive single-pass embeddings toward SEBI SCORES.
- **Agentic 2-Pass Routing:** (1) *Investigator Agent* identifies the product as a **Bank Fixed Deposit** (Commercial Bank entity). (2) Inspects negative modifiers: **SEBI SCORES Circular Para 5.3** explicitly excludes matters under RBI purview. (3) Executes targeted secondary search over **RBI Integrated Ombudsman Scheme 2021**: Clause 3(1)(j) (Bank = Regulated Entity) and Clause 9 (Interest shortfall = deficiency in service).
- **Audit & Grounded Determination:** *Verifier Agent* confirms all citations against extracted PDF text and enforces **Clause 10(2)(a)**: complainant must first lodge a 30-day written complaint to the bank before the Ombudsman accepts the filing.
- **Routing Decision:** refer_to_other_regulator → **Reserve Bank of India — Integrated Ombudsman Scheme (RB-IOS)**.

3. Verified Regulatory Corpus (SHA-256 Validated)

- **SEBI SCORES Master Circular** (SEBI/HO/OIAE/IGRD/CIR/P/2023/156): Para 1.2 (21 calendar-day resolution), Para 5.3 (statutory exclusions for RBI, IRDAI, PFRDA, CCI).
- **SEBI (Mutual Funds) Regulations, 1996** (amended to 01 Nov 2025): Reg 53 (Redemption transfer timelines), Reg 60A (21-day AMC grievance redressal).
- **RBI Integrated Ombudsman Scheme, 2021** (amended to 05 Aug 2022): Clause 3(1)(j) (Regulated Entity definition), Clause 9 (Grounds of Complaint), Clause 10(2)(a) (30-day prior bank complaint rule).

4. Key Requirement Verification & Metric Summary

Evaluation Metric / Component	Req ID	Target / Spec	Measured Result	Verification Outcome
Automated Test Suite	A1, D1, D5	High test coverage	**59 / 59 Passed**	Pytest suite passing clean
Golden Evaluation Batch	L3	≥ 20 scenarios	**21 / 21 Passed (100%)**	100% correct domain routing
Groundedness & Faithfulness	L3, F5	High fidelity RAG	**100.0% / 100.0%**	All citations resolve to PDF text
Hallucination Rate	L3	Zero tolerance	**0.0%**	Zero fabricated clauses / actions
Context Relevance	L1, L3	Capture modifiers	**100.0%**	Multi-pass captures Para 5.3 modifier
Guardrails & Safety	L4, F2	Block injections	**Verified**	Prompt injection & toxic inputs quarantined
Reliability & Circuit Breaker	A5	Graceful degradation	**Verified**	CLOSED → OPEN → HALF_OPEN transitions
Observability Stack	D5, A6	Full telemetry	**7/7 Targets UP**	Prometheus + 26-panel Grafana + Jaeger

5. Deployment & Quickstart Architecture

Runs completely under **Docker Compose** (7+ microservices: api, worker, redis, llm-stub, prometheus, grafana, jaeger):

- **Single Command Startup:** cp .env.example .env && docker compose up --build -d
- **Web UI & Grievance Portal:** http://localhost:8000 | **API Swagger:** http://localhost:8000/docs
- **Observability:** Prometheus at :9090, Grafana (26 panels) at :3000, Jaeger Tracing at :16686